Schwab ETF Trend Research

MACD Signals and Risk Review

Christina Winch consulting context | 2010

Art Ponomarev, PhD

Historical setting: mid-2010.
This setting and the research protocol are reconstruction assumptions.

Reconstructed 7 Nov 2026

Consulting context and provenance

Art's recollection

Art recalls supporting Christina Winch in 2010 with her Monday investment strategy and ETF risk/timing research. He had access to private records and worked alongside a colleague who programmed in C.

Public context, documented in 2011

An SEC filing distinguishes the stock-momentum Monday Portfolio from Moderate and Growth ETF portfolios. It supplies strategy context; role details follow Art's account.

Scope of this reconstruction

This ETF/MACD study uses synthetic examples and proposed methods. The original 2010 slide deck is unavailable; historical backtest results are not presented.

[5] Ginkgo Multi-Strategy Fund

Eight funds available by mid-2010

Ticker	Research exposure	First trading date
SCHB	U.S. broad market	3 Nov 2009
SCHX	U.S. large-cap	3 Nov 2009
SCHA	U.S. small-cap	3 Nov 2009
SCHF	International equity	3 Nov 2009
SCHG	U.S. large-cap growth	11 Dec 2009
SCHV	U.S. large-cap value	11 Dec 2009
SCHC	International small-cap	14 Jan 2010
SCHE	Emerging markets equity	14 Jan 2010

An available research universe, not eight independent portfolio exposures. The report defines inception as first secondary-market trading.

[1] Schwab Strategic Trust

Short histories constrain the study

About 5-8 months

Live ETF history at the illustrative cutoff of 30 June 2010.

- Indicator initialization consumes part of an already short observation period.
- The available sample cannot establish behavior across several market cycles.
- Earlier index or substitute-fund histories would be separate proxy studies, clearly identified as such.
- Later-launched funds and present-day fees must not be inserted into the assumed 2010 setting.

[1] Schwab Strategic Trust

A proposed research protocol

Question

Does a simple trend filter change exposure and drawdowns enough to justify its turnover and trading costs?

Baseline

Evaluate each ETF separately as a long/cash sleeve. Fix daily 12/26/9 MACD, an illustrative 60-session warm-up, and next-session-open execution before measuring results.

Validation

Compare a 90-session warm-up and nearby settings as sensitivity checks. Log every assumption and preserve chronological order; do not select the best-looking variant after the test.

No historical backtest or performance result is presented here.

MACD: one declared calculation

EMA(n, t) = alpha x Close(t)
+ (1 - alpha) x EMA(n, t-1)

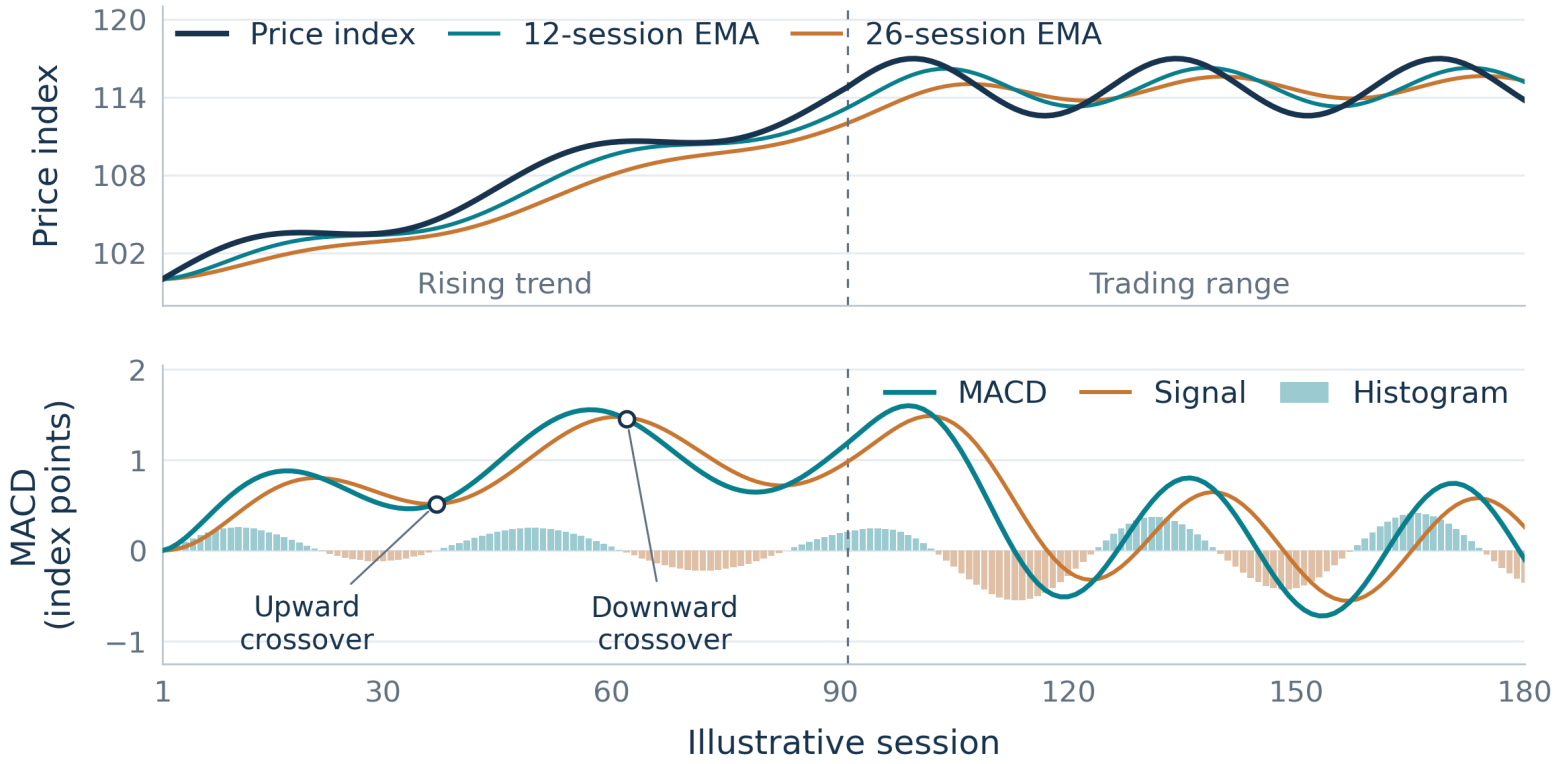
alpha = 2 / (n + 1)

MACD	EMA(12) - EMA(26)
Signal	EMA(9) of MACD
Histogram	MACD - signal

Initialize each recursive EMA from its first observation.
The synthetic figures use 160 hidden warm-up sessions; that is not the proposed live-data eligibility rule.

[2] Fidelity [3] Schwab

Crossovers: trend versus range

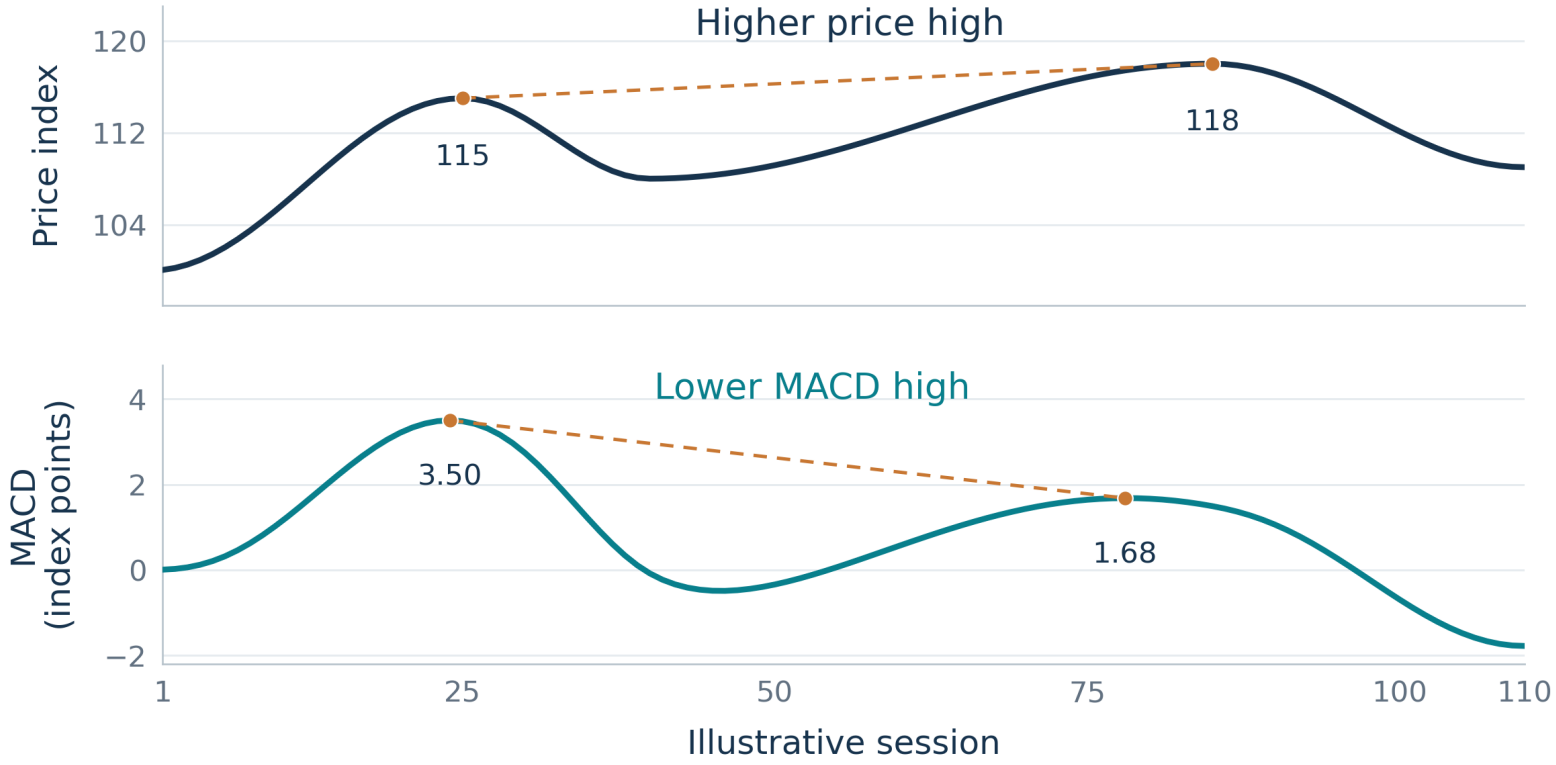


Synthetic series. No historical ETF prices or investment performance.

MACD/signal crossovers correspond to histogram zero crossings. Repeated crossings in a range can produce whipsaws.

[2] Fidelity

Bearish divergence: a warning

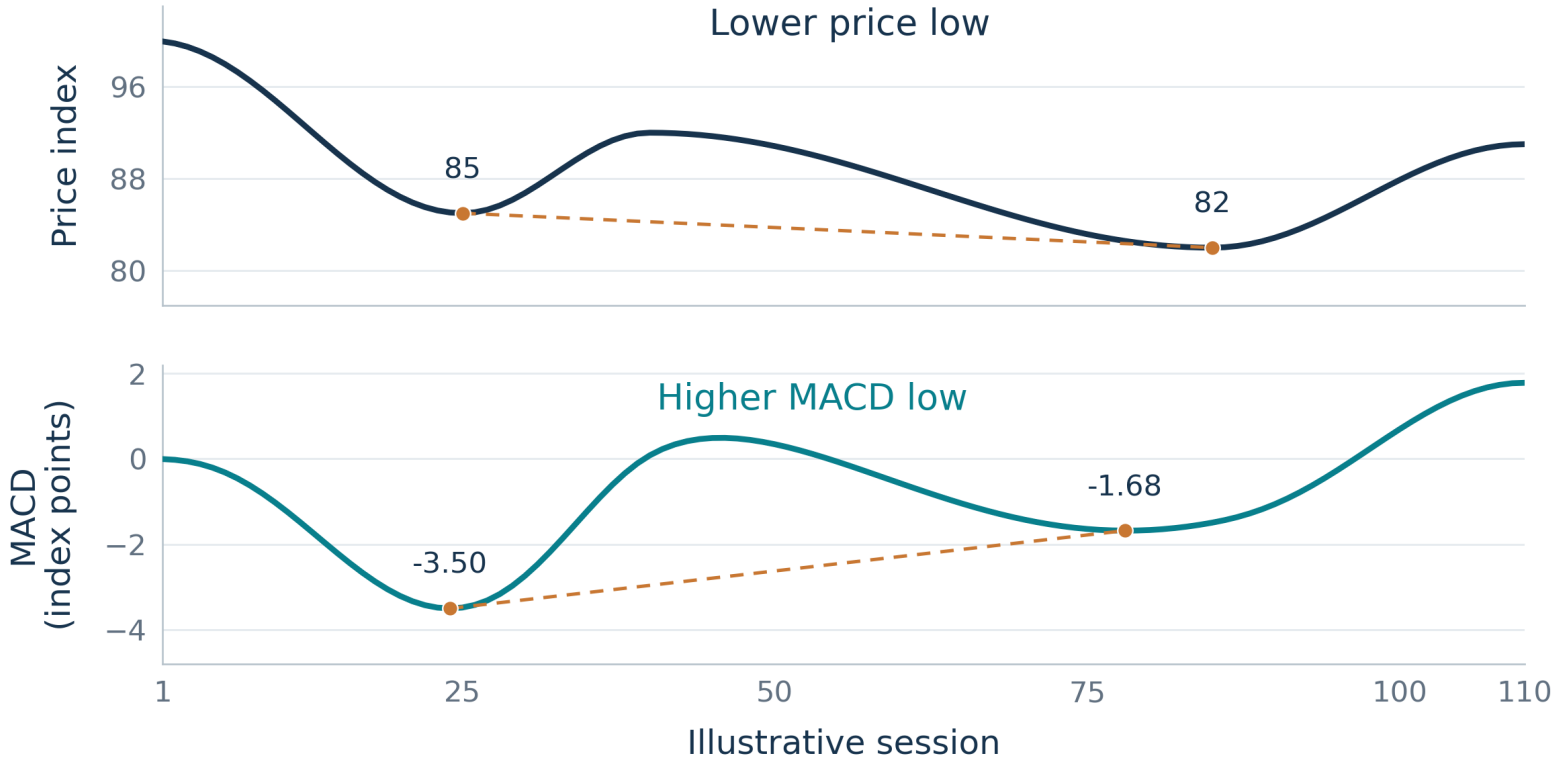


Synthetic series. No historical ETF prices or investment performance.

Higher price high, lower MACD high. Extrema need not coincide. Recognition requires a later confirmed pivot; reversal is not assured.

[4] Schwab

Bullish divergence: a warning



Synthetic series. No historical ETF prices or investment performance.

Lower price low, higher MACD low. The marked extrema are retrospective. Any systematic pivot rule must delay recognition until confirmation.

[4] Schwab

Proposed rule: long or cash

Let H(t) = MACD(t) - Signal(t). Start in cash.

Enter	When cash: H(t) > 0 and H(t-1) <= 0. Buy the sleeve.
Exit	When long: H(t) < 0 and H(t-1) >= 0. Sell to cash.
Otherwise	Keep the existing position, including when H(t) = 0.

Calculate after close t. Execute at the next available session's open t+1, with an explicit fill and cost model.

No shorting or leverage. Use the final warm-up observation as the prior bar; a positive first eligible value alone is not an entry.

Exposure and accounting must agree

- **Sleeves:** assign initial capital in advance. An inactive sleeve stays in cash. With no cross-sleeve rebalancing, portfolio weights may drift.
- **Execution:** old holdings earn the overnight gap until the next open. A new position does not earn the gap before its purchase.
- **Prices:** use consistent corporate-action adjustments, or raw fills with explicit split/dividend bookkeeping. Avoid double-counting dividends.
- **Costs and cash:** state commissions, spreads, slippage, and cash-return assumptions. Charge every modeled trade; do not subtract fund expenses twice.

Evaluation report and proposed outputs

Fair comparison

Use the same ETF and eligible dates for a buy-and-hold comparison. Apply consistent dividends, cash treatment, and realistic trading costs.

Metrics to calculate

Net return, drawdown, exposure/time invested, turnover, trade count, and cost sensitivity. A smaller drawdown with less market exposure does not alone demonstrate timing skill.

Proposed work products

A dated assumptions sheet, reproducible indicator calculations, an execution ledger, benchmark comparisons, and a short review of limitations.

Proposed deliverables; historical ETF metrics remain uncomputed.

Historical sources and provenance

[1] Schwab Strategic Trust, semiannual report

28 February 2010 - Supports fund identities and first secondary-market trading dates.

https://www.sec.gov/Archives/edgar/data/1454889/000095012310045321/f55658nvcsrs.htm

[5] Ginkgo Multi-Strategy Fund, SEC registration filing

1 April 2011 - Documents the Monday Portfolio name and its distinction from ETF portfolios; role details follow Art's account.

https://www.sec.gov/Archives/edgar/data/1314414/000091047211000504/coverpage.htm

Art's consulting role and private-record access are presented as his recollection. The illustrative ETF study is separate from the proprietary Monday strategy.

Indicator references and reading

[2] Fidelity, MACD indicator guide

Reference retrieved for this reconstruction - Standard MACD construction and crossover interpretation.

https://www.fidelity.com/learning-center/trading-investing/technical-analysis/technical-indicator-guide/macd

[3] Schwab, 3 Strength Indicators for Assessing Stock Momentum

6 June 2025 - Modern explanation of momentum indicators.

https://www.schwab.com/learn/story/3-strength-indicators-assessing-stock-momentum

[4] Schwab, Using Chart Divergences to Make Trading Decisions

9 September 2025 - Modern explanation of regular price/indicator divergence.

https://www.schwab.com/learn/story/using-chart-divergences-to-make-trading-decisions

The 2025 articles explain definitions for this 2026 reconstruction; they are not sources from the original 2010 work.